

**First Semester B.Tech (CBCS) Degree Examinations**

December 2024

MA1001-2 – MATRIX ALGEBRA AND CALCULUS

(For ACT, EC, EE, VLSI)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **One full** question from each **Unit**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- If A is a non-singular matrix of order n then rank of A is _____
 A) $n+1$ B) $n+2$
 C) less than n D) n
- The value of k for which the following equations $x + 2y - 3z = 6, 2x - y + 4z = 2, x + 2y + kz = 6$ have infinite solutions is _____
 A) 2 B) -3
 C) 3 D) 1
- If eigen values of a matrix A of order 3 are $\lambda_1, \lambda_2, \lambda_3$ then the eigen values of A^T are _____
 A) $\lambda_1, \lambda_2, \lambda_3$ B) $\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \frac{1}{\lambda_3}$
 C) $\lambda_1^2, \lambda_2^2, \lambda_3^2$ D) $\lambda_1^{-2}, \lambda_2^{-2}, \lambda_3^{-2}$
- Rank of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 3 \\ 0 & 4 & 0 \end{bmatrix}$ is _____
 A) 1 B) 2
 C) 3 D) 0
- $\frac{1}{1 \cdot 2^2} + \frac{1}{2 \cdot 3^2} + \frac{1}{3 \cdot 4^2} + \dots$ is _____
 A) Divergent B) Geometric series
 C) Convergent D) Arithmetic series
- The series $\frac{1}{1^p} + \frac{1}{2^p} + \frac{1}{3^p} + \dots$, converges if _____
 A) $p > 1$ B) $p < 1$
 C) $p \leq 1$ D) $p \geq 1$
- $\sum U_n$ is divergent if $\lim_{n \rightarrow \infty} (U_n)^{\frac{1}{n}} = l$ and l is _____
 A) $l > 1$ B) $l = 1$
 C) $l < 1$ D) $l \neq 0$
- The Maclaurin's series expansion of a function $f(x)$ is _____
 A) $f(1) + \frac{x}{1!} f'(1) + \frac{x^2}{2!} f''(1) + \frac{x^3}{3!} f'''(1) + \dots$ B) $1 + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots$
 C) $f(0) - \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) - \frac{x^3}{3!} f'''(0) + \dots$ D) $f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots$

9. The derivative of arc $\frac{ds}{dy}$ for the curve $x = f(y)$ is:
- A) $\sqrt{1 + \left(\frac{dy}{dx}\right)^2}$ B) $1 + \left(\frac{dy}{dx}\right)^2$
 C) $1 + \left(\frac{dx}{dy}\right)^2$ D) $\sqrt{1 + \left(\frac{dx}{dy}\right)^2}$
10. Two polar curves intersect orthogonally if,
- A) $\tan\phi_1 \cdot \tan\phi_2 = \frac{\pi}{2}$ B) $\tan\phi_1 \cdot \tan\phi_2 = 1$
 C) $\tan\phi_1 \cdot \tan\phi_2 = -1$ D) $\tan\phi_1 + \tan\phi_2 = 1$
11. The value of c got by applying Lagrange's mean value theorem to the function $f(x) = x^2$ in $[0, 4]$ is:
- A) 1 B) 2
 C) 3 D) 4
12. The radius of curvature of the curve $y = x^2$ when $x = 0$ is
- A) $\frac{1}{\sqrt{2}}$ B) $\sqrt{2}$
 C) 2 D) $\frac{1}{2}$
13. If $u = y^x$ then $\frac{\partial u}{\partial y}$ is:
- A) xy^{x-1} B) $y^x \log \log y$
 C) $y^x \log \log x$ D) y^x
14. If $u = x + y$ and $v = x - y$ the value of $J\left(\frac{u,v}{x,y}\right)$ is
- A) 2 B) 1
 C) -2 D) 0
15. If $f(x, y, z) = x^2 + xyz + z$, then the value of f_x at $(1, 1, 1)$
- A) 0 B) 3
 C) 1 D) -1
16. If $r = f_{xx}$, $s = f_{xy}$, $t = f_{yy}$, then $f(x, y)$ has maximum at (a, b) if,
- A) $rt - s^2 > 0$ & $r < 0$ at (a, b) B) $rt - s^2 < 0$ & $r < 0$ at (a, b)
 C) $rt - s^2 > 0$ & $r > 0$ at (a, b) D) $rt - s^2 < 0$ & $r > 0$ at (a, b)
17. Which of the following formula can be used to find area of a plane region D in the cartesian form?
- A) $A = \iint_D x \, dx \, dy$ B) $A = \iint_D xy \, dx \, dy$
 C) $A = \iint_D f(x, y) \, dx \, dy$ D) $A = \iint_D dx \, dy$
18. The value of $\int_0^{\frac{\pi}{2}} \sin^3 x \cos^4 x \, dx =$
- A) $\frac{3\pi}{280}$ B) $\frac{2}{3}$
 C) $\frac{2}{35}$ D) $\frac{35}{2}$

19. After changing the order of integration, the value of the integral $\int_0^a \int_y^a \frac{x}{x^2+y^2} dx dy$ is same as
- A) $\int_0^a \int_0^x \frac{x}{x^2+y^2} dy dx$ B) $\int_x^a \int_0^y \frac{x}{x^2+y^2} dy dx$
- C) $\int_0^a \int_x^a \frac{x}{x^2+y^2} dy dx$ D) $\int_x^a \int_y^a \frac{x}{x^2+y^2} dy dx$
20. Value of $\int_0^3 \int_1^2 \int_0^2 4 dx dy dz$ is
- A) 4 B) 24
- C) 48 D) 60

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

- | | Marks | BT* | CO* | PO* |
|---|-------|-----|-----|-----|
| 1. a) Test for consistency and solve the given system of equations by the Gauss elimination method.
$x + 4y + 9z = 16$
$2x + y + z = 10$
$3x + 2y + 3z = 18$ | 6 | L2 | 1 | 2 |
| b) Find the eigen values and eigen vectors of the matrix $\begin{bmatrix} 2 & 2 \\ 5 & -1 \end{bmatrix}$. | 6 | L3 | 1 | 2 |
| c) Find the rank of the following matrix using elementary row transformation.
$\begin{bmatrix} 1 & 3 & 1 & -2 \\ 1 & 4 & 3 & -1 \\ 2 & 3 & -4 & -7 \\ 3 & 8 & 1 & -7 \end{bmatrix}$ | 4 | L1 | 1 | 1 |
| 2. a) Solve the following system of equations using the Gauss-Seidel iteration method.
$2x + y - z = 12$
$2x - 8y + 5z = -5$
$-5x + 3y + 9z = -17$, start with $x^{(0)} = y^{(0)} = z^{(0)} = 0$ and carry out three iterations. | 6 | L3 | 1 | 2 |
| b) Using the Rayleigh's power method, find the dominant eigen value and the corresponding eigen vector of the matrix:
$\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$ with the initial approximation $[1 \ 0 \ 0]^T$. Carry out 4 iterations. | 6 | L2 | 1 | 2 |
| c) Check for consistency of the given system of equations.
$x + y + z = 6$
$x + 2y + 3z = 10$
$2x + 4y + 6z = 8$ | 4 | L1 | 1 | 1 |

Unit – II

- | | | | | |
|--|---|----|---|---|
| 3. a) State the D' Alembert's ratio test and hence discuss for convergence of the series $\left(\frac{1}{3}\right)^2 + \left(\frac{1 \cdot 2}{3 \cdot 5}\right)^2 + \left(\frac{1 \cdot 2 \cdot 3}{3 \cdot 5 \cdot 7}\right)^2 + \dots \infty$ | 6 | L2 | 2 | 1 |
| b) Check for convergence of the given series using the comparison test.
$\frac{\sqrt{2}-1}{3^3-1} + \frac{\sqrt{3}-1}{4^3-1} + \frac{\sqrt{4}-1}{5^3-1} + \dots \infty$. | 6 | L3 | 2 | 1 |
| c) Obtain the Maclaurin's series expansion of e^x up to fourth order derivatives. | 4 | L1 | 2 | 2 |

4. a) Obtain the Taylor's series expansion of $\sin x$ in powers of $\left(x - \frac{\pi}{2}\right)$ up to terms containing $\left(x - \frac{\pi}{2}\right)^4$. 6 L3 2 1
- b) Test for convergence of the given series using the D'Alembert's ratio test $\sum \frac{(n+3)!}{n! 3! 3^n}$. 6 L2 2 2
- d) Discuss for convergence of the given series using the Cauchy's root test. $\sum_{n=1}^{\infty} 3^n \left(\frac{n}{n+1}\right)^{n^2}$ 4 L1 2 1

Unit – III

5. a) Find the radius of curvature at any point on the cycloid $x = a(\theta + \sin\theta)$, $y = a(1 - \cos\theta)$. 6 L3 3 1
- b) Prove that with usual notation $\tan\phi = r \frac{d\theta}{dr}$. 6 L2 3 2
- c) Using the Lagrange's mean value theorem, find 'c' for $f(x) = \tan^{-1}x$ in $[0, 1]$. 4 L1 3 1
6. a) Find the angle of intersection between the curves $r = \frac{a}{1+\cos\theta}$ and $r = \frac{b}{1-\cos\theta}$. 6 L3 3 1
- b) State and prove the Cauchy's mean value theorem. 6 L2 3 1
- c) Using Lagrange's Mean value theorem find 'c' for $f(x) = \log x$ in $[1, e]$ 4 L1 3 2

Unit – IV

7. a) If $u = f(2x - 3y, 3y - 4z, 4z - 2x)$, then show that $\frac{1}{2} \frac{\partial u}{\partial x} + \frac{1}{3} \frac{\partial u}{\partial y} + \frac{1}{4} \frac{\partial u}{\partial z} = 0$. 6 L2 4 1
- b) Expand the function $f(x, y) = e^y \log(1+x)$ up to second degree terms using the Maclaurin's series expansion formula. 6 L3 4 2
- c) If $f = \log\left(\frac{x+y}{xy}\right)$, find the first order partial derivatives of f . 4 L1 4 1
8. a) If $u = \frac{yz}{x}$, $v = \frac{zx}{y}$, $w = \frac{xy}{z}$ then show that $\frac{\partial(u, v, w)}{\partial(x, y, z)} = 4$. 6 L2 4 1
- b) Find the extreme values of $x^2 + y^2 + 6x - 12$. 6 L3 4 2
- c) If $u = \frac{e^{x+y}}{e^x + e^y}$, then show that $u_x + u_y = u$. 4 L1 4 1

Unit – V

9. a) Prove that $\beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$. 6 L2 5 1
- b) Change the order of integration and hence evaluate $\int_0^{\infty} \int_x^{\infty} \frac{e^{-y}}{y} dy dx$. 6 L3 5 2
- c) Express $\int_0^{\infty} x^4 e^{-x^2} dx$ in terms of gamma function. 4 L1 5 1
10. a) Evaluate $\int_0^1 \int_1^2 \int_1^2 x^3 y z dx dy dz$. 6 L2 5 1
- b) Show that the area between the parabolas $y^2 = 4ax$ and $x^2 = 4ay$ is $\frac{16}{3}a^2$. 6 L3 5 2
- c) Evaluate $\int_0^2 \int_0^{x^2} x(x^2 + y^2) dy dx$. 4 L1 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
